

# Antonius Valentino

*Business Intelligence · Data Analyst*

Information Systems fresh graduate interested in building data-driven systems; from analytical pipelines and KPI layers to explainable ML models.

The following projects illustrate how I approach real business problems with a measurable technical approach.

## CORE STACK

Python

XGBoost · LightGBM

SHAP

Streamlit

Pandas · NumPy

Scikit-learn

Optuna

MySQL

Next.js

Ollama · Gemma 3

## DOMAIN

Credit Risk Modelling

Sports Analytics

Full-stack Development

Explainable AI

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2026

# Technology follows the problem

I view data as a tool for understanding business challenges and enabling better decision-making. Therefore, every project in this case study starts with real-world needs and problems faced by users or organizations, rather than being driven solely by an interest in a particular technology.

WORK APPROACH

Data leakage prevention

Every pipeline is designed with strict separation: the scaler is fit on the training data, transformed on validation/test data, and SMOTE is applied only to the training set. This is not just a best practice, but a way to ensure the reported metrics reflect model performance on unseen data.

Explainability as a core feature

From SHAP force plots to the Model Integrity page, every model decision must be explainable to a loan officer, a club analyst, or a laundry business owner.

Business metrics above model metrics

An AUC of 0.86 only becomes meaningful when translated into something concrete: how many defaults were successfully detected, how many manual working hours can be reduced, or what the impact is on daily business decisions.

TECHNICAL COMPETENCE

| AREA                | TOOLS                                                       | LEVEL                                                              |
|---------------------|-------------------------------------------------------------|--------------------------------------------------------------------|
| ML / Modeling       | XGBoost, LightGBM, Scikit-learn, Optuna, SMOTE              | <div><div></div><div></div><div></div><div></div><div></div></div> |
| Explainability      | SHAP (TreeExplainer, force plots, summary plots)            | <div><div></div><div></div><div></div><div></div><div></div></div> |
| Data Engineering    | Pandas, NumPy, Feature Engineering, Time-series split       | <div><div></div><div></div><div></div><div></div><div></div></div> |
| Visualization & App | Streamlit, Plotly, Matplotlib                               | <div><div></div><div></div><div></div><div></div><div></div></div> |
| Database & Backend  | MySQL, Next.js API Routes, Prepared Statements              | <div><div></div><div></div><div></div><div></div><div></div></div> |
| Business Analysis   | KPI design, business metric interpretation, domain research | <div><div></div><div></div><div></div><div></div><div></div></div> |

PENDIDIKAN

BSc Information Systems  
Telkom University · 2021–2026

# Three projects, three different domains

Each project was built from scratch, from data exploration to deployment as an interactive application or production system.

|                                                                                                                           |                                                                                                                                   |             |                            |               |  |
|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------|---------------|--|
| <div>01</div> <div>Predictive Credit Scoring for Indonesian MSMEs</div> <div>Fintech · Credit Risk · Explainable AI</div> |                                                                                                                                   |             | XGBOOSTSHAPSTREAMLITOPTUNA |               |  |
| PROBLEM                                                                                                                   | SOLUTION                                                                                                                          | KEY METRICS |                            |               |  |
| Manual, slow, and inconsistent MSME credit evaluation. No system explains <i>why</i> a borrower is rejected.              | End-to-end ML pipeline: EDA, Feature Engineering, XGBoost, Optuna, SHAP, up to a Streamlit app with a threshold override feature. | 86.4%       | 55%                        | <1s           |  |
|                                                                                                                           |                                                                                                                                   | AUC-ROC     | DEFAULT DETECTED           | DECISION TIME |  |

|                                                                                                                                           |                                                                                                                    |                 |                              |  |  |
|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------|--|--|
| <div>02</div> <div>xFlair — Predicting Nutmegs in the 2026 World Cup</div> <div>Sports Analytics · Predictive Modelling · StatsBomb</div> |                                                                                                                    |                 | XGBOOSTSTATSBOMBPYSHAPPLOTLY |  |  |
| PROBLEM                                                                                                                                   | SOLUTION                                                                                                           | KEY METRICS     |                              |  |  |
| No system quantitatively measures a football player's <i>flair</i> and creativity from event data.                                        | Time-series pipeline from 522K StatsBomb events, using a Player-Match formulation to avoid mathematical tautology. | 27.9%           | 522K                         |  |  |
|                                                                                                                                           |                                                                                                                    | MAE IMPROVEMENT | EVENTS PROCESSED             |  |  |
|                                                                                                                                           |                                                                                                                    | 0               |                              |  |  |
|                                                                                                                                           |                                                                                                                    | DATA LEAKAGE    |                              |  |  |

|                                                                                                                                                                 |                                                                                                                                     |                 |                                       |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------------------------------|--|--|
| <div>03</div> <div>WHSER — Laundry Management Platform</div> <div>Full-stack · Business Intelligence · LLM Integration</div>                                    |                                                                                                                                     |                 | NEXT.JSMYSQLSELF-HOSTED LLMTYPESCRIPT |  |  |
| PROBLEM                                                                                                                                                         | SOLUTION                                                                                                                            | KEY METRICS     |                                       |  |  |
| Unstructured high-volume laundry operations: orders are easily missed, SLAs are frequently violated, and there is no visibility into profitability per service. | Dual-layer architecture: operational layer (real-time POS) plus analytical layer (immutable snapshot plus 8 KPIs plus LLM insight). | 20              | 8                                     |  |  |
|                                                                                                                                                                 |                                                                                                                                     | DATABASE TABLES | KPIS CALCULATED                       |  |  |
|                                                                                                                                                                 |                                                                                                                                     | Est. 30h        |                                       |  |  |
|                                                                                                                                                                 |                                                                                                                                     | SAVED/WEEK      |                                       |  |  |

# Predictive Credit Scoring for Indonesian MSMEs

PROBLEM STATEMENT

Banks and fintechs serving Indonesian MSMEs face a fundamental dilemma: the manual credit evaluation process can take 30 to 60 minutes per application, results are inconsistent across loan officers, and there is no explanation provided to customers when their applications are rejected. As a result, two problems arise simultaneously: risky borrowers slipping through the selection, and eligible borrowers being rejected

The core challenge is not just building an accurate model, but **optimizing the decision threshold**; the boundary between "approve" and "reject" that reflects the business risk appetite, not just statistical metrics.

BUSINESS IMPACT

86.4%

AUC-ROC TEST SET

55%

DEFAULT DETECTED

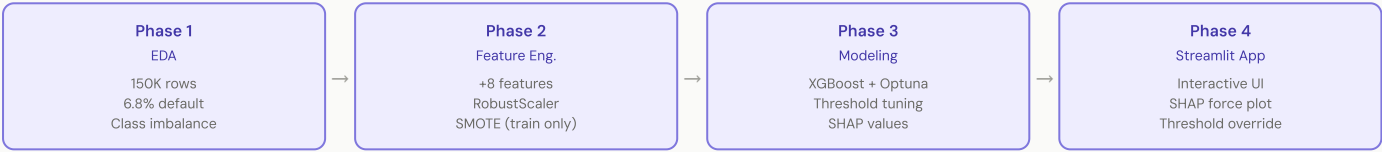
<1s

DECISION TIME

\$374k

ESTIMATED SAVINGS/YEAR

END-TO-END PIPELINE



WHAT SETS THIS PROJECT APART

Threshold as a Policy Artifact

The 0.5354 threshold is saved as a separate JSON file, version-controlled, and can be overridden from the UI without retraining the model.

Zero Data Leakage by Design

RobustScaler is fit only on the training set. SMOTE is applied only to the training set. Validation and test sets reflect the true production distribution

Explainability is Mandatory, Not Optional

Every credit decision is accompanied by a SHAP force plot. A loan officer can read specific reasons, such as "two 90-day late payment incidents pushed the score toward default."

# Critical technical decisions and their reasons

01 — RobustScaler vs StandardScaler

DATA PREP

**Problem:** The data has extreme outliers (income up to \$3.8M, utilization 1.5). StandardScaler uses mean and standard deviation, which are sensitive to outliers.

**Decision:** RobustScaler uses the median and IQR, making it more resistant to outliers without sacrificing the interpretability of the linear transformation.

02 — Stratified Split (60/20/20)

VALIDATION

**Problem:** The target class is imbalanced (93.2% good, 6.8% default). A random split risks producing an uneven distribution.

**Decision:** `train_test_split(..., stratify=y)` ensures each split has approximately 6.8% default, resulting in comparable metrics.

03 — SMOTE only on Train Set

IMBALANCE

**Problem:** Applying SMOTE before the split causes data leakage, as synthetic samples from the minority class will "leak" into the validation set.

**Decision:** SMOTE is fit and transformed only on the training set. Validation and test sets remain reflective of the imbalanced production distribution.

04 — Threshold Tuning via F1 Max

BUSINESS LOGIC

**Problem:** A 0.5 threshold is an arbitrary number that does not reflect asymmetric costs: a false negative (approving a defaulter) is far more expensive than a false positive (rejecting a good borrower).

**Decision:** Sweep the threshold on the validation set and select the one that maximizes the F1-minority score. Result: threshold 0.5354, saved as `best_threshold.json`.

05 — SHAP vs LIME

EXPLAINABILITY

**Problem:** Loan officers and regulators require accountable explanations, including for audit or regulatory compliance purposes.

**Decision:** SHAP uses Shapley values from game theory, which is theoretically stronger than LIME. It supports both local (per borrower) and global (feature importance across the entire dataset) explanations.

MODEL COMPARISON

| MODEL               | AUC-ROC | F1-MINORITY | STATUS                           |
|---------------------|---------|-------------|----------------------------------|
| XGBoost (Tuned)     | 0.8641  | 0.4421      | <div>Selected</div>              |
| LightGBM            | 0.8503  | 0.3735      | +1.4% AUC but -15.7% F1-minority |
| Logistic Regression | 0.8475  | 0.3184      | Baseline                         |

XGBoost was chosen not because it had the highest AUC, but because its **F1-minority score was 38% higher than LightGBM**. In credit risk, detecting defaults is the primary business priority, not inference speed.

OPTUNA HYPERPARAMETER TUNING

| PARAMETER        | BEST VALUE | SEARCH RANGE |
|------------------|------------|--------------|
| n_estimators     | 350        | 100–500      |
| max_depth        | 5          | 3–8          |
| learning_rate    | 0.08       | 0.01–0.3     |
| scale_pos_weight | 3.2        | 1–5          |

PERSISTED RTIFACT

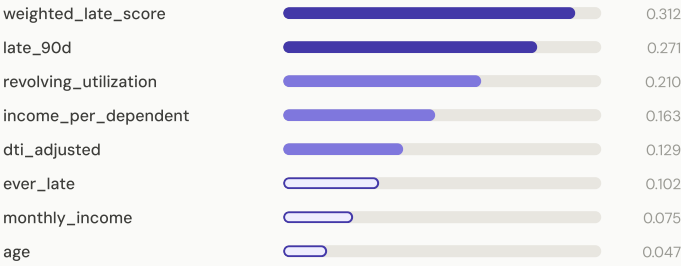
`best_xgb_model.json` — XGBoost model serialized

`scaler.pkl` — RobustScaler for inference

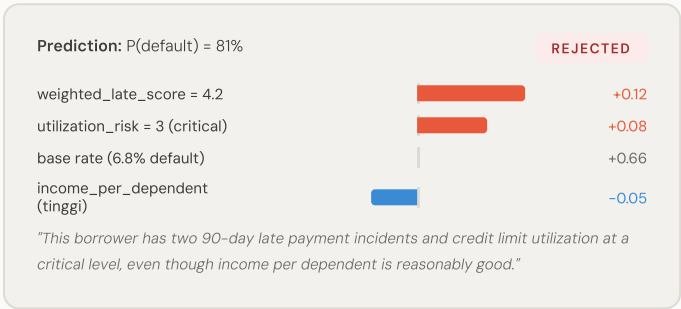
`best_threshold.json` — Decision boundary (0.5354)

# Every credit decision must be explainable

SHAP FEATURE IMPORTANCE (GLOBAL)



SHAP FORCE PLOT — BORROWER EXAMPLE (LOCAL)



THRESHOLD AS A POLICY INSTRUMENT

The threshold is a business decision. By saving it as a separate artifact, the lending team can simulate policy changes without touching the model code.

| THRESHOLD | APPROVAL RATE | EST. DEFAULT RATE | STRATEGY           |
|-----------|---------------|-------------------|--------------------|
| 0.30      | ~40%          | <2%               | Ultra-conservative |
| 0.5354    | ~70%          | ~5-6%             | Balanced (tuned)   |
| 0.70      | ~95%          | ~15%              | Highly inclusive   |

FEATURE ENGINEERING — BUSINESS RATIONALE

- weighted\_late\_score**

`late_30 × 1 + late_60 × 2 + late_90 × 4` - 90-day late payments are a charge-off signal (banking industry standard). The 4× weight reflects a different level of severity.
- income\_per\_dependent**

An income of 5 million with 4 dependents means something very different from 5 million with no dependents. This feature captures family financial pressure that is not visible from raw income numbers alone.
- dti\_adjusted**

Debt-to-income ratio is a banking regulatory standard (safe limit 30-40%), sso this feature allows the model to "speak the same language" as the risk manager.

# What I learned and its limitations

## CRITICAL LEARNINGS

### 1. "Perfect" results are a red flag

In early iterations, the model MAE was almost zero. I was skeptical, repeated the process, and found that the target variable could be mathematically derived from two features (tautology). This taught me that honest validation is more important than impressive metrics.

### 2. The threshold is a business decision, not a technical choice

I initially used a 0.5 threshold by default. After understanding the context of MSME credit more deeply, I realized the right threshold must reflect the asymmetric costs between false negatives and false positives, and must be adjustable without retraining.

### 3. Feature engineering from domain knowledge over statistical correlation

Adding `weighted_late_score` (which stems from the understanding that 90-day late payments are the charge-off limit in banking) provided more meaningful improvement than feature engineering based solely on correlation.

## ACKNOWLEDGED LIMITATIONS

- **Data is from 2009** (financial crisis) — so the default rate might not represent current Indonesian MSME conditions.
- **Recall is only 55%** — meaning approximately 45% of defaults slip through the prediction. The model needs to be combined with manual review for borderline cases.
- **Fairness audit has not been conducted** — age as a feature has the potential to become a proxy for discrimination. A disparate impact audit is required before production deployment.

## PROJECT SUMMARY

*"Developed explainable credit risk model (86.4% AUC-ROC) with interactive policy dashboard; detected 55% of defaults pre-approval through XGBoost + SHAP pipeline with zero data leakage; operationalized decision threshold as version-controlled artifact enabling risk appetite simulation without model retraining"*

## DESIGN DECISION

### Trade-off yang menentukan pilihan model:

*"In this project, I chose XGBoost not because it had the highest AUC, but because its F1-minority score was 38% better than LightGBM. In credit scoring, a model that misses defaults is a business failure, even if its AUC is perfect"*

*This insight came from understanding that the cost of a false negative, approving a borrower who later defaults, is far greater than the cost of a false positive, rejecting an actually eligible borrower. That is why I also saved the threshold as a separate artifact that the lending team can adjust without needing to call a data scientist."*

## LINK

GitHub: [github.com/val-Antonius/predictive\\_credit\\_scoring\\_for\\_umkm](https://github.com/val-Antonius/predictive_credit_scoring_for_umkm)

Demo App: [shap-credit-scoring.streamlit.app](https://shap-credit-scoring.streamlit.app)

# 02 xFlair – Nutmeg, measured with data

PROBLEM STATEMENT

A nutmeg, passing the ball between an opponent's legs, is not a high-level skill move. Any player can do it. But only a small fraction choose to do it under match pressure. That is what makes it interesting: it is not a performance indicator, but rather a signal of *boldness* and *flair* that is purely for entertainment.

Sports media, fantasy sports platforms, and the broadcast industry have no systematic way to measure this. The question of this project is **whether we can predict how many times a player will attempt a nutmeg before the match even starts?**

BUSINESS VALUE

522K  
EVENTS PROCESSED

27.9%  
MAE IMPROVEMENT

Use case: fan engagement & broadcast

Players with a high predicted nutmeg rate are highly sellable content for pre-match narratives, fantasy sports, or exotic betting markets.

PIPELINE ARCHITECTURE



Why Nutmeg?

Precisely because a nutmeg is not a performance indicator, it becomes an interesting proxy for flair. It is purely about boldness and creativity in a 1v1 moment. From an analytical perspective, predicting an event that is "unnecessary but chosen" is much harder than predicting an event that is actually planned.

Model Integrity Page

Not many portfolios transparently display their model limitations. xFlair has a dedicated page for a decile calibration chart and error analysis, building trust rather than hiding weaknesses.

Target Audience

Sports media analysts, fantasy sports platforms, and the broadcast industry. A nutmeg is an entertainment event, and this project attempts to measure something that has previously only lived in highlight reels.



# Finding & solving Mathematical Tautology

## THE TAUTOLOGY PROBLEM

### First Iteration – It looked successful.

The first model with player-level aggregation produced a very low MAE. The predictions looked perfect; all players received a score around 0.71.

This was a red flag. After investigation, it was found that the target variable could be mathematically derived from two features:

$$y = \text{nutmegs} / \text{total\_dribbles} = f(\text{feature\_1} / \text{feature\_2})$$

The model was not learning patterns; it was just doing arithmetic. This was a mathematical tautology.

## REDESIGN: PLAYER-MATCH FORMULATION

### BEFORE (PLAYER-LEVEL)

Each row = one player  
Target = career nutmeg rate  
Feature = career stats  
Problem:  $y = f(x_1/x_2)$



### AFTER (PLAYER-MATCH)

Each row = one match  
Target = nutmeg in match T  
Feature = data before match T  
Solution:: Predict the future from the past

## SHIFT(1) — TIME-SERIES INTEGRITY

Rolling features use `shift(1)` — meaning each row only uses information from previous matches, not the match being predicted.

```
rolling_nutmeg_rate_last5 = shift(1).rolling(5).mean()
```

First match for each player: rolling feature equals 0 (no prior history). This is methodologically honest.

## FEATURES USED

| FEATURE                   | DESCRIPTION                       | SOURCE      |
|---------------------------|-----------------------------------|-------------|
| career_nutmeg_rate        | Nutmeg / total dribble historical | Rolling     |
| rolling_nutmeg_rate_last5 | Average of last 5 matches         | shift(1)    |
| rolling_dribbles_last5    | Exposure / past opportunity       | shift(1)    |
| opponent_duel_win_rate    | Opponent defensive strength       | StatsBomb   |
| career_dribble_success    | General dribbling skill           | Aggregation |
| match_number              | Experience in this tournament     | Derived     |

## MODEL RESULT

0.306

MAE (MODEL)

0.424

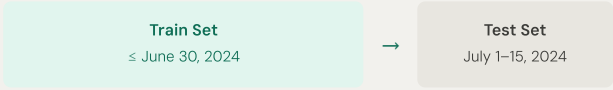
MAE (BASELINE)

27.9%

IMPROVEMENT

## TIME-SERIES SPLIT — METHODOLOGICAL INTEGRITY

Random splitting is forbidden for time-series data. Leaking future data into the training set is one of the most dangerous forms of data leakage.



### Design Decision

*"Results that are too good are a danger sign, not a success. Perfect results must be investigated further, not immediately celebrated."*

# The Flair Index – from event data to per-player predictions

## THE FLAIR INDEX — TOP PLAYER RANKING

EXPECTED NUTMEG / MATCH · WC 2026 PREDICTION

|              |                |
|--------------|----------------|
| Neymar Jr.   | 0.87 predicted |
| Richarlison  | 0.72 predicted |
| Vinícius Jr. | 0.65 predicted |
| Mbappé       | 0.58 predicted |
| Salah        | 0.43 predicted |

\*Player names are illustrative for format demonstration.

## MODEL INTEGRITY — ACTIVE TRANSPARENCY

### Main Differentiator of This Project

The Model Integrity page explicitly displays:

1. Decile calibration chart (predicted vs actual)
2. Error analysis per player
3. Dataset Limitations(only 3 tournaments)

*"Transparency builds trust that cannot be built even with high accuracy."*

## PROJECT SUMMARY

*"Built end-to-end sports analytics ML pipeline predicting nutmeg frequency for WC 2026 flair players using XGBoost on 522,885 StatsBomb events; achieved 27.9% MAE improvement over baseline with zero data leakage via Player-Match time-series formulation."*

## KEY LEARNINGS

### Data granularity determines model validity

Player-level aggregation looks simpler, but it results in tautology. The Player-Match formulation is more complex, but it is the only methodologically valid way for time-series prediction.

### A small dataset is not a roadblock, but a design challenge

664 rows is a very small dataset for ML. The solution is not more data (since data is limited), but stricter regularization and more informative features.

### Acknowledged limitations build greater trust

The Model Integrity page actually makes this project more convincing. Displaying weaknesses openly shows that I understand the model's limitations, not just how to build it.

## LIMITATIONS & DEVELOPMENT

- Currently only covers 3 international tournaments; domestic league data has not been used
- Players without history in these 3 competitions cannot yet be predicted.
- Dataset of 664 rows (currently being developed by expanding data coverage to further validate results)

## LINK

GitHub: [github.com/val-Antonius/xflair](https://github.com/val-Antonius/xflair)

Demo App: [xflair.streamlit.app](https://xflair.streamlit.app)

# WHSER - From manual operations to a controlled system

PROBLEM STATEMENT

High-volume laundry businesses rely on spreadsheets, handwritten notes, and phone calls. As a result: orders are lost, SLAs are violated without early detection, inventory is mismanaged, and owners lack visibility into profitability per service type.

WHSER transforms laundry operations from unstructured manual workflows into a controlled, proactive system through a **dual-layer architecture**: an operational layer for day-to-day staff workflows, and an analytics layer for the owner's strategic decisions.

BUSINESS IMPACT

100%

ORDER TRACKED

30h

SAVED / WEEK

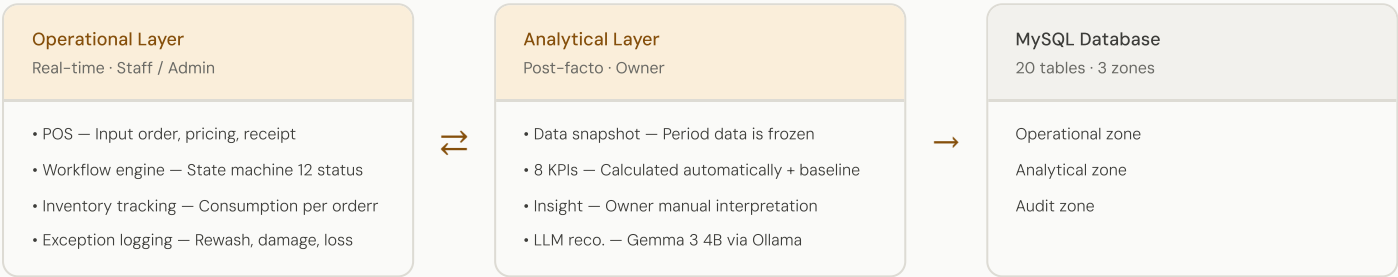
20

DATABASE TABLES

50+

API ENDPOINTS

DUAL-LAYER ARCHITECTURE



Order as the Single Source of Truth

All data (items, status history, exceptions, inventory consumption) is linked to a single entity: the orders table. Every status change is immutably recorded in the order\_status\_log.

Immutable Snapshot for Financials

analytical data is frozen at the period boundary, so a January report will not change even if data is updated in February. This is crucial for financial audit consistency and historical decision-making.

LLM as Augmentation, Not Automation

Recommendations from Gemma 3 require owner approval before execution. This human-in-the-loop approach ensures business decisions remain in the right hands.

# From transaction events to strategic decisions

PIPELINE: RAW DATA → INSIGHT



8 CALCULATED KPIS

| KPI                  | SHORT FORMULA                  | TARGET   | SIGNIFICANCE |
|----------------------|--------------------------------|----------|--------------|
| SLA Compliance       | on_time / total_completed      | >95%     | Normal       |
| Rewash Rate          | rewashes / completed           | <5%      | Attention    |
| Exception Freq.      | exceptions / total orders      | <2%      | Critical     |
| Contribution Margin  | revenue - material - labor     | >70%     | Normal       |
| Inventory Variance   | (actual - expected) / expected | <10%     | Attention    |
| Aging Distribution   | count(>72h) / total            | <10%     | Normal       |
| Productivity         | completed / days_in_period     | Baseline | Normal       |
| Capacity Utilization | peak_concurrent / capacity     | 60–80%   | Normal       |

CRITICAL TECHNICAL DECISIONS

- Separation of Operationa & Analytical**

Analytical queries running on live data cause database lock contention and slow down transactions. Snapshots separate this load: transactions remain fast, and analytics do not compete with the POS
- Manual Snapshot, Not Automated**

The owner decides when to analyze (weekly, monthly, or ad-hoc). Automated snapshots would create too much unused data. Agency remains in the hands of the business owner.
- MySQL Without ORM**

Prepared statements provide full control over queries, better performance for prototypes, and built-in prevention of SQL injection. An ORM adds unnecessary abstraction at this phase.
- Role-Based Dual Dashboard**

Admins see real-time workflows. Owners see strategic trends. A single dashboard trying to serve both can create cognitive overload, making two dashboards with different contexts the more appropriate choice.

TECH STACK

NEXT.JS 14

TYPESCRIPT

MYSQL

OLLAMA

GEMMA 3 4B

JSPDF

REACT

# Systems thinking behind every design decision

## LLM INTEGRATION — IMPLEMENTATION DETAILS

## Human-in-the-Loop Pipeline

- 1 Owner notes insights from the KPI dashboard (human interpretation, not automated)
- 2 Gemma 3 reads insights + metrics to generate specific operational recommendations
- 3 Owner reviews & approves relevant recommendations (human gate)
- 4 Tasks are created and assigned to staff, effectiveness is measured in the next snapshot

## CRITICAL LEARNINGS

## Good architecture solves business problems before technical problems

The separation of operational and analytical layers was born from the understanding that owners and staff have fundamentally different information needs, not from technical preference.

## Immutability is a feature, not a limitation

An unchangeable snapshot might look like a limitation at first. But in the context of financial audits and decision accountability, immutability is a requirement, not a trade-off.

## Manual-first is a choice, not a shortcoming

Inventory consumption is manually input by staff, not via IoT sensors. This is not a budget limitation, but a design decision: the system must be deployable today, in any laundry, without specialized hardware.

## PROJECT SUMMARY

*"Architected dual-layer laundry management system separating operational (real-time POS) from analytical (immutable snapshot-based analytics) concerns; computed 8 business KPIs with baseline comparison and significance detection; integrated local LLM (Gemma 3) for human-approved operational recommendations."*

## DECISION DESIGN

## Choosing the right solution, not the most complex one:

*"I briefly considered real-time analytics with data streaming. It is technically more impressive. But after understanding the business needs, a laundry owner does not need real-time data every second. An immutable snapshot is the right answer"*

## DEVELOPMENT STATUS

|                                |           |
|--------------------------------|-----------|
| Phase 1: Operational App       | Completed |
| Phase 2: Role-based Dashboards | Completed |
| Phase 3: Analytics & Snapshots | Completed |
| Phase 4: LLM Integration       | Completed |
| Phase 5: Task Management       | Completed |

## LINK

GitHub: [github.com/val-Antonius/whser](https://github.com/val-Antonius/whser)

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GLAD YOU MADE IT TO THE END.

# I'd love to discuss *any of these projects further*

These three projects reflect how I approach problems: starting from a real business question, building solutions that can be explained, and making sure every technical decision has a reason behind it.

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